QECTOR Decoder Workbench

QUICK START GUIDE

Application version 3.5.1
Decoder backend qector-decoder-v3 0.6.9 (bundled; minimum supported 0.6.2)
47-tool MCP server, 13 decoders, 9 code families

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1. Install and run in one minute

Every edition is distributed without the decoder wheel embedded. On first launch the application downloads and installs qector-decoder-v3 from PyPI automatically. After that it works offline.

Windows

- Double-click QectorWorkbench-Portable.exe. That is all; it is a single file. First launch downloads the decoder backend from PyPI.
- Or run the installer QectorWorkbenchSetup.exe for a Start-menu and desktop icon.

Linux

- sudo dpkg -i ./qector-workbench_3.5.1_amd64.deb
- sudo apt-get -f install (only if dpkg reports missing dependencies)
- Launch from the Science menu or run: qector-workbench

macOS

- Open QectorWorkbench.dmg and drag the app to Applications.
- Right-click the app the first time and choose Open. First launch downloads the decoder backend from PyPI.

2. Your first decode

- Open the Code Explorer tab. Choose rotated_surface, set distance 3, and click Build.
- Open the Decoder Lab tab. Choose the decoder auto_router, set a physical error rate (for example 0.05) and a seed, and run.
- Read the result: syndrome_valid should be true, with a correction weight and the logical outcome. Change the seed to see a reproducible new shot.
- Try the Benchmark tab to sweep many shots, or Batch and Streaming for bulk decode.

3. Use it from an AI agent (MCP)

The same application is a headless MCP server. Start it with the --mcp flag and connect any MCP client:

```
Windows :  QectorWorkbench-Portable.exe --mcp
Linux   :  qector-workbench --mcp
macOS   :  /Applications/QectorWorkbench.app/Contents/MacOS/QectorWorkbench --mcp
```

See the QECTOR MCP Integration Guide for client configuration and tool examples.

4. Good to know

- 13 decoders and 9 code families are built in.

- auto_router is a safe default: it routes each code to a suitable decoder automatically.
- The qLDPC families (bicycle, bivariate_bicycle) need bp_osd or a routed decoder, not union-find.
- All benchmark numbers are simulation outputs; regenerate them on your hardware.